

30**A**

$$Bqv = qE$$

$$v = E / B = 1400 / (2.0 \times 10^{-3}) = 7.0 \times 10^5 \text{ m s}^{-1}$$

$$hf = \phi + \frac{1}{2}mv^2$$

$$\phi = \frac{hc}{\lambda} - \frac{1}{2}mv^2 = \frac{(6.63 \times 10^{-34})(3 \times 10^8)}{450 \times 10^{-9}} - \frac{1}{2}(9.1 \times 10^{-31})(7.0 \times 10^5)^2$$

$$\phi = 2.2 \times 10^{-19} \text{ J}$$

~ THE END ~